Fulton–Harris pilot
Detailed formalization plan

Lectures 1–2, items P01–P12, and the exact S_3 example

12 October 2026; hierarchy and F2 update 20 October 2026
companion to `docs/fulton-harris-plan.tex`

What this document adds. The whole-book plan fixes the pilot's *scope*: 12 items from Lectures 1–2, the S_3 regression example and a choice-free policy. This document turns that scope into work: concrete data structures, proposed Arend signatures, a proof route for each item, the infrastructure lemmas each route needs, a dependency-ordered schedule, and acceptance commands. The original library survey used the checkout on 12 October 2026 with the rebuilt CLI (`-ss`, `-fu`, `-ch`). Signatures marked *sketch* are design proposals that have not been typechecked yet.

Summary

- **The M0 blocker is resolved for the pilot cone.** A rebuilt `cli-1.12.0-full.jar` typechecks all 447 modules. A fresh `-no-daemon -r -c` check of the six representation modules exits 0 with no errors (Section 1).
- **The pilot cone is axiom-free by imports.** No representation, linear-algebra, trace, tensor, rational, complex or symmetric-group module can reach `Logic.Classical` or `Logic.Choice`. In the whole library, only `Arith/Complex/FTA/Classical.ard` uses an axiom, and nothing imports it.
- **The pilot uses a descendant class.** `FinLinRepres` inherits finite freeness from `FinModule`; `FinIrreducibleRepres` adds a nonzero-vector witness and the split invariant-submodule dichotomy. The tested submodule needs no basis. Schur and character theorems take the descendant directly. The old standalone `IsFinIrreducible` predicate has been removed. The general-module design remains D3b.
- **The pilot adds linear-algebra infrastructure.** Starting from the existing bases, coordinates, matrices and Smith-form results, it adds basis-independent traces, coordinates for sums, Hom and tensor products, kernel/image splittings under their stated hypotheses, and the averaging trace formulas. The plan records these as L01–L22 (L08 split into L08a and L08b); the implementation status is recorded below.
- **Decidable zero only where it is used.** The definitions are stated over `Ring`, `CRing` or `Field`. `DiscreteField` is a hypothesis only of the kernel and image constructions that test pivots: image splitting (L08b), Schur via kernel and image (P03), and dimension of invariants (P11). Averaging stays over `CRing` with `Monoid.Inv (natCoef card)`.
- **A small constructive-$\mathbb{C}$ bridge is in the pilot.** The explicit S_3 example is transported to `ComplexField` (R8a and R8b). This covers its matrices, decomposition maps, character values and scalar commutant, with explicit inverses for the denominators 2, 3 and 6. Irreducibility over $\mathbb{C}$ gets its own argument, through the scalar commutant and Maschke averaging (L21), which needs no zero test. General scalar-extension machinery stays deferred.
- **One deliberate deviation from the book.** P07 (uniqueness of multiplicities) goes through characters, so it comes after P12. It assumes characteristic 0 through injectivity of `natCoef`, which holds over $\mathbb{Q}$, and it inherits the dependence of P12 on Exercise 1.2. Both are recorded: the relation is `restricted` and the status is `conditional-on-exercises`.

1 Baseline established on 12 October

Check	Result	Status vs. 8 Oct
Rebuild	`./gradlew :cli:jarDep :arend-lib:meta:classes` succeeds; `cli/build/libs/cli-1.12.0-full.jar`; OpenJDK 21.0.10.	Was: unsuitable jar
Whole library	Daemon bootstrap typechecked and serialized all 447 modules in 44.7 s with no `[ERROR]` (loads existing `.arc` where valid).	New
Fresh pilot cone	`arend -no-daemon -r -c` on `Irreducible`, `MaschkeLemma`, `Characters`, `PermutationRepresentation`, `Product`, `LinRepresCat`: exit 0, zero errors, 10.8 s.	Was: parse failure at `Equiv:15:24`
Axioms present	`\axiom` s: `alem`, `achoice` (`Logic/Classical.ard:9,11`); `adchoice`, `acchoice` (`Logic/Choice.ard:6,26`). `alem` has no usages.	Confirmed
Axiom reachability	Import closure (over-approximation) of every pilot module avoids both axiom modules. Only importer of `Logic.Choice`: `Arith/Complex/FTA/Classical.ard:18`, itself unimported.	New; sound because every answer is "no"
Goal reporting	Named goals parse: `{?name}` and `{?name(e)}` (`Arend.g4:262`). Printed as `[GOAL] ... Goal name`. Goals do *not* set a failing exit code; `-c` treats GOAL like ERROR.	New
Daemon caveat	A daemon fixes `-c`, `-r`, `-serialize`, `-L` at start-up and ignores per-request values. Acceptance runs must pass `-no-daemon`.	New
Tests	`test/` has 34 meta/solver tests and no representation tests. Negative tests use the `fails` meta.	Confirmed

Remaining M0 items. (1) Run `arend -no-daemon -r -c` and `-t -r -c` over the whole library, and record the hashes of the jar, `arend-lib/ext` classes and the source tree. (2) Add the two control tests of Section 8. (3) Check in the import-closure audit script under `tools/autoformalization/`. (4) Note one stale comment: `Arith/Complex/FTA.ard:42` names a nonexistent `Arith.Complex.FTA.Choice`.

2 Pilot scope and exit criteria

In scope.

- P01–P12 from the whole-book plan.
- One extra item, P01b: an Arend proof that the old irreducibility notion implies excluded middle.
- The S_3 regression suite: R1–R7 over $\mathbb{Q}$, and R8a/R8b, the explicit transport of the example to constructive $\mathbb{C}$ (structure, then characters).
- The infrastructure L01–L22 that these need.

Out of scope; recorded as deferred.

- General scalar-extension machinery: transfer of arbitrary theorems from $\mathbb{Q}$ to $\mathbb{C}$ (an M2 bridge).
- The general irreducibility notion for arbitrary, possibly infinite-dimensional, modules (D3b).
- The exterior-square conjunct of Prop. 2.1.
- A general decomposition algorithm for Cor. 1.6.
- Completeness of the S_3 character table: that the number of irreducibles equals the number of classes.

- Hermitian forms.
- Proofs of Exercises 1.2 and 2.5, which stay as intentional holes.

Exit criteria. There are two tiers, because the tensor basis (L14) is allowed to finish after everything else.

1. **Core pilot complete.**
   (a) Every P-, L- and R-item has a manifest record whose `statement_review` has a recorded outcome.
   (b) A fresh `-no-daemon -r -c` run of the touched modules reports exactly two goals, `FH_L01_Ex1_2` and `FH_L02_Ex2_5`, and no unregistered goals or errors.
   (c) L14 and the tensor conjunct of P09 may be absent from the source, but their manifest records must say `pending` with the reason. No goal may stand in for them.
   (d) Every declaration outside the dependency closure of the two exercise goals is reported `checked`. Every declaration inside it is reported `conditional-on-exercises`.
   (e) The axiom audit reports no path to `Logic.Classical` or `Logic.Choice`.
   (f) The effort measurements of Section 10 are recorded.
2. **All selected items complete.** The core criteria hold, and in addition L14 and the tensor conjunct of P09 are checked, so no item is `pending`.

Only the second tier is reported as "pilot complete" without qualification.

3 Design decisions

D1. Scalars

- Group-theoretic identities (actions, traces of products, averaging) are stated over `{R : CRing}`. Averaging takes `cardInv : Monoid.Inv (R.natCoef G.card)`, an explicit inverse passed to the standalone function `mean_func`.
- Definitions use the weakest class that makes them meaningful. The split subrepresentation works over any `Ring`, and `FinIrreducibleRepres` over any `NonZeroRing`, which excludes the zero ring. The Hom and dual representations need a `CRing`.
- `DiscreteField` is a hypothesis only of constructions that test pivots, that is, kernel and image bases built from `image-fin`, `kernel-fin` and `smith-bases` (`Algebra/Linear/VectorSpace.ard:125--187`). These need a `SmithDomain`; `-ch` confirms that `DiscreteField` is one. See L08 for which of these split over a general `SmithDomain`.
- Results whose proofs avoid pivots are stated over `Field` or `LocalRing`, so that they apply to `ComplexField`. Examples are L21 and the R8a/R8b bridge.
- Concrete examples use `RatField : DiscreteOrderedField Rat` (`Arith/Rat.ard:145`).
- For $|G|^{-1}$ over $\mathbb{Q}$, build the inverse from `RatField.nonZero-Inv` and `natRat/=0` (`Arith/Rat.ard:337`). This is infrastructure lemma L16.

D2. Finite-dimensional representations: truncated versus chosen bases

The finite interface is implemented by inheritance. Computation uses explicit arrays and basis proofs. The optional `BasedLinRepres` wrapper below remains a design sketch; the implemented constructors take its data as separate arguments.

```
-- Algebra/Group/Representation/Finite.ard
-- Parent order preserves the conventional scalar/group argument order.
\class FinLinRepres \extends FinModule, LinRepres     -- isFinModule : ∃ basis (truncated)

-- sketch: optional wrapper for a chosen basis
\class BasedLinRepres \extends LinRepres {
  | basis : Array E
  | isBasis : IsBasis basis
}
\func BasedLinRepres.toFin (B : BasedLinRepres) : FinLinRepres  -- forget the choice
```

Rules:

- Scalars and dimensions over a `FinLinRepres` are defined only through basis-independence. This follows the `dimension-pair` pattern: a `\level \Sigma` over the truncated basis (`Algebra/Module/FinModule.ard:47`).
- Constructions such as matrices, splittings and averaged projectors take explicit basis arguments; they do not require that optional wrapper.
- No global choice of a basis is made anywhere.

D3. Finite-dimensional irreducibility via split invariant subspaces (pilot notion)

The existing `SubLRepres` is the canonical membership-based interface: its predicate P defines the subtype $W = \{v : E \mid P(v)\}$ with inclusion $j(v, p) = v$. Its descendant `SplitSubLRepres` adds a *linear* retraction $r : E \to W$ satisfying $rj = \mathrm{id}_W$. No basis of W is assumed. The retraction need not commute with the action. The module classes follow the same hierarchy; coordinates are added only in the based descendants. The following excerpt shows the implemented class structure; the displayed ellipsis stands for the unchanged coordinate laws.

```
-- Algebra/Module/SubLModule/Split.ard
\class SplitSubLModule \extends SubLModule {
  | retract : LinearMap S (IModule
      (\new SubLModule { | SubLModule => \this }))
  | retract_incl (x : IModule
      (\new SubLModule { | SubLModule => \this }))
      : retract x.1 = x
}

-- Algebra/Linear/BasedSplitSubLModule.ard
\class BasedSplitSubLModule \extends SplitSubLModule {
  | dim : Nat
  | basisIncl : LinearMap (ArrayFinModule {R} dim) S
  | coords : LinearMap S (ArrayFinModule {R} dim)
  ...
}

-- Algebra/Group/Representation/SplitSubLRepres.ard
\class SplitSubLRepres \extends SplitSubLModule, SubLRepres {
  \override S : LinRepres R G
  -- sub is the canonical membership subtype ILinRepres
  -- incl is its canonical inclusion; retr is retract
}
\class BasedSplitSubLRepres
  \extends BasedSplitSubLModule, SplitSubLRepres {
  \override S : LinRepres R G
}

-- Algebra/Group/Representation/Irreducible.ard
\class FinIrreducibleRepres.{u} \extends FinLinRepres.{u} {
  \override R : NonZeroRing.{u}
  \override G : Group.{u}
  \override E : \Set u
  | nontrivial : ∃ (v : E) (v /= 0)
  | irreducible (S : SplitSubLRepres R G
      (\new LinRepres { | LinRepres => \this }))
      : (\Pi (x : S.sub) -> x = S.sub.zro) || IsSurj S.incl
}
```

The base classes register `SubLModule.toLModule` and `SubLRepres.toLinRepres` as coercions to the structures on their members. For a subobject `W`, `x : W` denotes a member and `W v` is its membership proposition; `W.S` remains the ambient object. The split and based descendants inherit the coercions. The explicit instances `IModule` and `ILinRepres` call these same constructions, so the module and representation coercions agree on the underlying module.

The irreducibility condition considers an invariant submodule $W \subseteq E$ with a supplied linear map $r : E \to W$ satisfying $r(w) = w$ for every $w \in W$. It requires no basis of W. A finite basis is assumed to exist for the ambient representation E.

One may also specify the subrepresentation using a separate representation U, an intertwiner $j : U \to E$, and a linear map $s : E \to U$ satisfying $s(j(u)) = u$. The lemma `splitMono-irreducible` then concludes that U is zero or j is surjective. Its proof applies the irreducibility condition to the image $W = j(U)$. The map $j : U \to W$ is an isomorphism, with inverse $w \mapsto s(w)$; no basis of U is used. The functions `fromSplitMono` and `imageEquiv`, together with `splitMono-dichotomy`, implement this proof.

A basis of W is additional data, recorded by the based descendants. The maps `coordinateIso` and `arrayToSub` relate those coordinates to the vectors of W.

- **Linear, not equivariant, retraction.** A split monomorphism in `RepresentationCat` is an invariant direct summand, and testing only those defines indecomposability. Over $\mathbb{F}_2$, the swap action of C_2 on $\mathbb{F}_2^2$ is indecomposable but not irreducible: the line spanned by $(1,1)$ is invariant and linearly split, but has no invariant complement. The equivariant split is what averaging produces (`maschke-split`, P04a), when $|G|$ is invertible.
- **Universe levels.** The membership and split classes inherit their universes from their parent classes and ambient module. The finite irreducible descendant retains its explicit parameter `.\{u\}`. Categorical conclusions such as `maschke-split` use `RepresentationCat` at that level.
- **No decidable equality in the definition.** Finite freeness is inherited from `FinModule`. The `nontrivial` field asserts $\exists v \in E,\ v \neq 0$; the lemma `basis-nonempty` proves that any basis is nonempty. The `irreducible` field expresses the split-submodule dichotomy. No basis is chosen, and no `dimension` or decision procedure is needed. This descendant class replaces the former standalone predicate and its anonymous pair of proofs.
- **Scalar and module nontriviality.** The class retains the `NonZeroRing` coefficient convention. Its `nontrivial` field separately excludes the zero representation, by witnessing a vector $v \neq 0$. Both $0 \neq 1$ and $v \neq 0$ are negated equalities, not positive apartness or invertibility evidence. No zero test is assumed. The nonzero-vector field is essential: finite freeness alone includes the zero module. `DiscreteField` only enters the constructions that produce image splittings (L08b, P03).
- **Why this escapes the W_P argument.** A split subspace comes with a retraction. For $W_P \subseteq \mathbb{Q}$ that would mean a linear $r : \mathbb{Q} \to W_P$ with $r \circ j = \mathrm{id}$, so $j \circ r$ is multiplication by an idempotent $c \in \mathbb{Q}$, and $c = 0$ or $c = 1$ decidably. Building that data already settles P.
- **Invariance is a proposition.** It is an equation in a set, so no data has to be chosen to state it.
- **Recorded restrictions.**
  - The notion only makes sense for finite free representations, and it tests split invariant submodules without requiring a basis of the submodule.
  - It is the pilot's working definition, not the general replacement. Arbitrary and infinite-dimensional modules (for example Verma modules) need the separate design item D3b, which is out of scope here.
  - The name `IsIrreducible` stays unassigned until D3b is decided.
- **Open questions for M1 review.**
  - Resolved on 20 October: `nontrivial` is a nonzero-vector witness; finite freeness is inherited, and nonemptiness of the basis is derived.
  - `sub` is an arbitrary representation, so finite projective summands are allowed. Rank-one irreducibility then needs a ring without nontrivial idempotents: over $R \times R$, the summand $R \times 0$ is a split subrepresentation. Is this the right generality for D3b?
- **The old definition.** It is renamed `IsStronglyIrreducible` and kept together with its Schur proof, so that P01b can state the excluded-middle consequence. The doc comment says the notion is classical-strength and must not be used as a default.

D4. Representations built from others

```
-- sketch; Algebra/Group/Representation/Hom.ard
\func HomLRepres {R : CRing} {G : Group} (A B : LinRepres R G) : LinRepres R G
  -- carrier LinearMapLModule A B;  (g . phi) v = g ** phi (inverse g ** v)
\func DualLRepres {R : CRing} {G : Group} (A : LinRepres R G) : LinRepres R G
  => HomLRepres A (TrivialAction (RingLModule R) G)
-- Algebra/Group/Representation/Tensor.ard
\func TensorLRepres {R : CRing} {G : Group} (A B : LinRepres R G) : LinRepres R G
  -- carrier TensorProductModule A B; g acts by tensorLift of (a, b) |-> g a ⊗ g b
```

- The direct sum is the existing `ProductLRepres`.
- Finite indexed sums are iterated `ProductLRepres`. They are only needed in the S_3 example and in P07, both with a small fixed number of summands.
- The library's misspelled `InterwiningMap` (and the module of that name) was renamed to `IntertwiningMap`, overriding the whole-book plan's instruction to keep the existing spelling.

D5. Characters

- The existing `Character E bv g` becomes the *based* character.
- The basis-free `character` on a `FinLinRepres` is defined by eliminating the truncated basis into a level-(-1) Σ-type, as `dimension-pair` does. This needs L05.
- Class functions are `G -> R` with `\Pi g h -> f (h * g * inverse h) = f g`.
- The pairing is the inverse-character one: $\langle \chi, \psi \rangle = |G|^{-1} \sum_g \chi(g^{-1})\psi(g)$.
- Comparing with the book's conjugate pairing belongs to the deferred $\mathbb{C}$ bridge.

D6. Exercise holes

Exercises 1.2 and 2.5 are stated in full. Their bodies are the named goals `{?FH_L01_Ex1_2}` and `{?FH_L02_Ex2_5}`, and each declaration's doc comment cites the exercise. Consumers are written against the statement and inherit `conditional-on-exercises`.

- **To be checked in M0.** The checker must still elaborate a consumer whose dependency has a goal body.
- **Fallback.** If it cannot, the consumer takes the exercise statement as an explicit hypothesis. Its name gets the suffix `-cond`; for example, the conditional form of `orthogonality` is `orthogonality-cond`.

D7. Placement and names

New declarations go in mathematical modules under `Algebra/Group/Representation/` and `Algebra/Module/`, and they follow the naming table of the `arend-formalize` skill. Book numbers appear only in doc comments (`- | Fulton-Harris Prop.~1.5`) and in the manifest. The S_3 objects that are reusable (the sign representation and the standard representation of S_n for $n = 3$) go in the library. The S_3 numerical checks go in `test/Algebra/Representation/S3Test.ard`.

4 Infrastructure backlog

Each L-item is a lemma or construction missing from the checkout. Its absence was confirmed by symbol search, usages and signature search. Sizes are rough proof-effort classes: S is under 30 lines, M is 30–150, L is over 150.

ID	Statement (sketch)	Reuse	Size	Consumers
L01	`toLinearMap_*`: `E.toLinearMap (g * h) = E.toLinearMap g ∘ E.toLinearMap h`	`**-assoc`, `exts`	S	L05, P08, P11
L02	Matrices of the action reverse products: `toMatrix lv bv (toLinearMap (g*h)) = toMatrix .. h product toMatrix .. g`. So $g \mapsto$ matrix of $\rho(g)$ is an *anti*homomorphism into `MatrixRing`; the homomorphism is $g \mapsto$ matrix of $\rho(g^{-1})$, or a map from the opposite group	L01, `toMatrix_*` (`LinearMap.ard:109`; row convention)	S	P01, P08, P09
L03	`toMatrix` is linear in the map: `toMatrix_+`, `toMatrix_*c`, `toMatrix_BigSum`, `toMatrix_FinSum`	`basis-split-unique`, `linearMap_BigSum`	M	P11
L04	Trace is basis independent: `Trace (toMatrix lu bu f) = Trace (toMatrix lu' bu' f)` for an endomorphism	`change-basis_matrix-left`, `-right`, `toMatrixInv`, `basis-iso`, `Trace-conjugation` or `Trace-prod`	M	L05
L05	`endoTrace {R : NonZeroCRing} (U : FinModule R) (f : LinearMap U U) : R`, defined by the `dimension-pair` pattern, with `endoTrace-char`	L04	S	P08, P11
L06	Trace of a block-diagonal matrix: `Trace (blockMatrix A B) = Trace A + Trace B`	`blockMatrix`, `elem00/11` (`Matrix.ard:51`)	S	P09
L07	Basis of `ProductLModule`: `in1`-images followed by `in2`-images; matrix of a product map is `blockMatrix`	`IsBasis`, `basis-split-unique`	M	P09
L08a	**Kernel splits over a `SmithDomain`.** With `smith-bases`, $f(u'_j) = d_j v'_j$, and $\ker f$ is spanned by the u'_j with $d_j = 0$, because a `SmithDomain` is a domain and $d_j = 0$ is decidable. Projection onto those coordinates is a retraction.	`smith-bases`, `kernel-fin`	M	P03, P04b
L08b	**Image splits only with invertible pivots.** Over a general `SmithDomain` it fails: $2\cdot : \mathbb{Z} \to \mathbb{Z}$ has image $2\mathbb{Z}$, which has a basis but no retraction from $\mathbb{Z}$. So the image of f is split if every nonzero Smith coefficient d_j is invertible; this holds automatically over a `DiscreteField`. The retraction sends $v'_j \mapsto d_j^{-1} e_j$ for $j <$ rank.	`smith-bases`, `image-fin`	M	P03

ID	Statement (sketch)	Reuse	Size	Consumers
L09	Over a `DiscreteField`: `retr ∘ incl = id` and `dim = T.dim` (a full split subspace) imply `IsSurj incl`	`toMatrix_*`, `matirx-inv` (sic, `Matrix.ard:542`), `matrix-split_<=`	S	P03
L10	Dimension bound: a split inclusion has `dim <= T.dim` over a `NonZeroCRing`	`matrix-split_<=` (`Matrix.ard:645`)	S	L21, R3
L11	Basis of `LinearMapLModule U V` for based U, V: matrix units; upgrade `matrix-equiv` (set-level, `LinearMap.ard:134`) to a `ModuleIso`	L03	M	P02, P12
L12	Kronecker product of matrices and `Trace (kron A B) = Trace A * Trace B`	`BigSum` double-sum lemmas	M	P09, P12
L13	Matrix of $\varphi \mapsto b \circ \varphi \circ a$ in the matrix-unit basis is `kron (transpose A) B`	L11, L12, `Trace-transpose`	M	P12
L14	Basis of `TensorProductModule` of based modules $(e_i \otimes f_j)$; tensor of linear maps; matrix is a Kronecker product	`tensorLift`, `tensorLift-unique`	L	P09
L15	Trace of an idempotent: for $p \circ p = p$ on a based module over `DiscreteField`, `endoTrace p = natCoef (dimension (image-fin p))`	L08, L07, `image-fin.dimension_rank`	M	P11
L16	`Inv (RatField.natCoef (suc n))`; injectivity of `natCoef` over `RatField`	`nonZero-Inv`, `natRat/=0`	S	P04, P07, R*
L17	`SymFinGroup (n : Nat) : FinGroup (Sym n)` from `SymmetricGroup` and `SymFin` (no instance exists today); check that the `DecSet` fields agree and that `EquivFin` does not win instance search	`Algebra/Group/Symmetric.ard: 65,284,297`	S	R1–R7
L18	Natural action `SymAction n : GSet (SymmetricGroup n)` on `Fin n`. Products are left-to-right (`(e * e') j = e' (e j)`, `Equiv.ard:233`), so the action must be `e ** i = inverse e i`; then `PermRepr` gives $(g \cdot f)(j) = f(g\,j)$	`symQEquiv`, `Sym.equals`	S	R2, R4, R5
L19	Fixed points as a decidable subset of a finite `GSet` with decidable equality, with a `FinSet` instance (`SigmaFin.DecSubSet-isFin`); conjugacy classes of an explicit finite group as decidable subsets	`DecSubSet`, `SigmaFin`, `FinSum-inj`	S	P10, R1
L20	Inversion $g \mapsto g^{-1}$ as a `QEquiv` for every `Group`, and the reindexing `FinSum (x o inverse) = FinSum x` (today only `SymmetricGroup.inv-isEquiv` exists)	`inverse-isInv`, `FinSum_Equiv`	S	P09, P11, P12

ID	Statement (sketch)	Reuse	Size	Consumers
L21	**Irreducibility from a scalar commutant.** Over a commutative `LocalRing` R (hence `NonZeroRing`), with `cardInv`: if $E \cong R^n$ with $n > 0$ (a full split subspace T with `0 < T.dim`) and every intertwiner $E \to E$ is scalar, then return a `FinIrreducibleRepres` extending E. Positive rank is essential, since the zero representation also has only scalar endomorphisms. Route: a split invariant S gives the equivariant idempotent $\pi = \mathit{incl} \circ \rho$ (P04a). By the commutant $\pi = c \cdot \mathrm{id}_E$, so $(c^2 - c)\,\mathrm{id}_E = 0$. Evaluating at $T.\mathit{incl}(e_0)$ and applying $T.\mathit{retr}$ gives $(c^2 - c)e_0 = 0$ in R^n, hence $c^2 = c$; this step uses $n > 0$. Then `LocalRing.connected` (`Algebra/Ring/Local.ard:37`) gives $c = 0$ (so the submodule is zero, using $0 \neq 1$) or $c = 1$ (so `incl` is surjective). No zero test is used, so it applies to `ComplexField`.	P04a, `LocalRing.connected`, L10	M	R3, R8a
L22	Ring homomorphism $\mathbb{Q} \to \mathbb{C}$: `fromRealRingHom` (`Arith/Complex.ard:108`) composed with the structure map of `RealField : OrderedFieldAlgebra RatField Real`; transport of matrices (`matrix-map`), of traces, and of `Monoid.Inv` along a ring homomorphism, which gives the inverses of $2, 3, 6$ in `ComplexField`	`fromRealRingHom`, `RingHom`, `charPoly_map` pattern	S	R8a, R8b

Notes.

- L14 is the largest item. The pilot needs the tensor character formula, so L14 is scheduled early, in wave 1, as an independent track. If it stalls, only the tensor conjunct of P09 is held up; P12 uses the Hom route (L11–L13) instead.
- For L14, `Algebra/Algebra/TensorProduct.ard` builds a bilinear multiplication by nesting `tensorLift`. That is the template for the tensor $f \otimes g$ of two linear maps. Note that `tensorLift_tensor` is propositional, not `idp`.
- L03 can reuse `AddMonoidHom.func-FinSum` (`Algebra/Monoid/MonoidHom.ard:92`), which applies to linear maps.
- L08 is the design-critical item, and it is split in two because the image half is false over a general `SmithDomain` (the counterexample $2\mathbb{Z} \subseteq \mathbb{Z}$ is in L08b). It is prototyped and typechecked first, together with D3 and P03.
- The image of an *idempotent* is always split, with π itself as the retraction onto its image. So P04, P11 and L21 never need L08b.

5 Item specifications

Each item lists the source statement, the constructive target, the relation to the source (`same`, `reformulated`, `restricted`), the proposed declarations, the proof route, dependencies and acceptance. Page numbers are printed pages; the physical PDF page is the printed page plus 15, an offset checked on pp. 3–16.

P01 Representations, based finite representations, split subobjects, irreducibility

Source. §1.1, pp. 3–4. A representation is a homomorphism $\rho : G \to \mathrm{GL}(V)$. V is irreducible if it has no proper nonzero invariant subspace.

Target.

- D2 (`FinLinRepres` and `BasedLinRepres`).
- D3 (`SplitSubLRepres` and the pilot class `FinIrreducibleRepres`).
- The bridge to $G \to \mathrm{GL}(V)$, with the order fixed. `LinearMapRing` multiplies as `f * g = g âĹŸ f` (`Algebra/Module/Instances.ard:28`), while an action satisfies $\rho(gh) = \rho(g) \circ \rho(h)$ (L01). So $g \mapsto \rho(g)$ is an *anti*homomorphism into the units of `LinearMapRing E`. The bridge is therefore `LinRepres.toGroupHom : GroupHom G (units of LinearMapRing E)` with $g \mapsto \rho(g^{-1})$. The equivalent alternative, a homomorphism from the opposite group with $g \mapsto \rho(g)$, is recorded for review. Either way, the construction comes with its inverse.

Relation: `reformulated`. Invariant subspaces are replaced by split ones.

Route.

- The induced action on R^{dim} is $g \cdot x = retr(g \cdot incl\, x)$.
- Its `**-assoc` uses `incl-invariant` and `retr_incl`.
- The bridge reuses `toLinearMap`, `toLinearMap_ide`, L01 and `inverse_*`. A regression test pins the direction of the homomorphism.

Migration.

- Restate `PermutationRepresReducible.Not-Irreducible` (`PermutationRepresentation.ard:57--67`) for the new notion, over an arbitrary ring and an arbitrary `GSet X`, with the same hypotheses as now. The old theorem's conclusion `0 = {R} 1` turns into the hypothesis that R is a `NonZeroRing`, and the new conclusion is `Not (FinIrreducibleRepres { | LinRepres => PermRepr X })`.
- The constant line $j(c) = (c, c, \dots)$ with $r(f) = f(x_0)$ is a split invariant subspace of dimension 1 for any X. `nonConstantVector` shows that j is not surjective. The refutation uses only the `irreducible` field of `FinIrreducibleRepres`.
- **Recorded change of strength.** For infinite X, the new class requirement of finite freeness generally fails anyway. The migrated theorem is then weaker than the old one, which refuted irreducibility in the strong sense for every X. The general-module form waits for D3b.
- `SubLRepres.IsTrivial` stays, documented as the strong dichotomy.

Acceptance. One-dimensional examples are irreducible over any `NonZeroCRing`: the trivial and sign representations (L10 gives $dim \leq 1$). The migrated `Not-Irreducible` checks.

P01b The old notion implies excluded middle

Target. `IsStronglyIrreducible.excludedMiddle`.

From `IsStronglyIrreducible (TrivialAction (RingLModule R) G)` and $0 \neq 1$ in R, derive `\Pi (P : \Prop ) -> P || Not P`.

Route.

- The submodule is W_P with `contains x => P || x = 0`. Its closure fields are short case splits.
- Then follow the two cases of the whole-book plan, page 4.
- This is an ordinary constructive theorem. It introduces no axiom and documents why D3 is needed.

Size. S. It is a good first task for calibrating the workflow.

P02 Direct sum, dual and Hom representations; Exercise 1.2 (hole)

Source. §1.1, pp. 4–5; Exercise 1.2.

Target. `HomLRepres` and `DualLRepres` (D4); the existing `ProductLRepres`. Exercise 1.2 as an intentional hole:

```
-- | Fulton--Harris Exercise 1.2: invariant vectors of Hom(V, W) are the G-linear maps.
\lemma invariantHom {R : CRing} {G : Group} {A B : LinRepres R G} (f : LinearMap A B)
  : (\Pi (g : G) -> g HomLRepres.** f = f) <-> (\Pi (g : G) (a : A) -> f (g ** a) = g ** f a)
  => {?FH_L01_Ex1_2}
```

Relation: `same`.

Route. The action fields of `HomLRepres` are pointwise, using `exts`, `**-assoc` and `inverse_*`. The pattern to follow is `IntertwiningMapLModule` in `LinRepresCat.ard`.

Acceptance. `HomLRepres` and `DualLRepres` check, and only the registered goal remains.

P03 Schur's lemma (1)

Source. Schur 1.7(1), p. 7.

Target.

```
\lemma SchurLemma.{u} {K : DiscreteField.{u}} {G : Group.{u}}
  {A B : FinIrreducibleRepres.{u} K G} (f : IntertwiningMap A B)
  : f.IsZero || IsEquiv f
```

Relation: `reformulated`. It uses the pilot irreducibility. As implemented in wave 1 it concludes `IsEquiv f`, like the old lemma, rather than `ModuleIso`; `ModuleIso.iso<->inj+surj` converts between the two. The `DiscreteField` hypothesis is needed for the image splitting (L08b), not for the definitions.

Route.

- The kernel of f is split by L08a, which works over any `SmithDomain`. The image is split by L08b, which needs invertible pivots, so this is where `DiscreteField` is used. The target is a proposition, so the truncation in `smith-bases` can be eliminated.
- Invariance comes from `KernelLRepres` and `ImageLRepres`.
- Apply the named `irreducible` fields of the domain and codomain to these split presentations. The zero/whole cases give the conclusion directly; no dimension comparison is required.

Risk. This item validates D3, so it is prototyped first, typechecked together with L08a and L08b. If the image splitting cannot be stated cleanly, revise D3 before writing anything else.

Not covered. Schur over constructive $\mathbb{C}$. Specific complex maps can have explicitly known kernel and image bases. What is missing is a general procedure that constructs such bases, and their retractions, without additional evidence such as invertible pivots. So the $\mathbb{C}$ side of the pilot uses L21 instead.

P04 Maschke: invariant complements

Source. Prop. 1.5, p. 6.

Target. Two declarations.

- `P04a` over `CRing`, with $|G|^{-1}$. Given a split invariant subspace S, it returns an equivariant idempotent $\pi = \mathit{incl} \circ \rho$, where ρ is the averaged retraction, with $\pi \circ \mathit{incl} = \mathit{incl}$. Its image is S, and it needs no basis and no zero test, so it also applies to `ComplexField`. L21 uses it.
- `P04b` over `DiscreteField`. It returns a complement S' as a `SplitSubLRepres` together with an isomorphism of representations $E \cong S \oplus S'$.

Relation: `restricted`. The input is split rather than an arbitrary subspace; the whole-book plan, page 6, explains why.

Route.

- Average $p_0 = \mathit{incl} \circ \mathit{retr}$. Reuse `SumOverGroup`, `FinSumRewrite` and the `mean-func-preserve` argument (`Group/Representation/MaschkeLemma.ard`), generalized from a `SubLRepres` target to `E`.
- The complement is $\ker \pi$, split by $\mathrm{id} - \pi$. L08a supplies its basis over any `SmithDomain`; `DiscreteField` is stated for uniformity with P03, and weakening it is a review experiment.
- The isomorphism is `coprod-map` of the two inclusions, with inverse $(\pi, \mathrm{id} - \pi)$.

Regression. Characteristic $p \mid |G|$: over `FinField` with $p = 2$ and $G = S_3$, `cardInv` is unavailable. A test documents that the hypothesis is needed.

P05 Complete reducibility for the presented inputs

Source. Cor. 1.6, p. 6.

Target. An explicit decomposition of the permutation representation of S_3 on $\mathbb{Q}^3$ as the trivial line plus the standard representation. It is an isomorphism of representations with both summands proved irreducible. Relation: `restricted`; the general corollary is deferred.

Route. Apply P04b to the split line $(1, 1, 1)$, or write both inclusions out explicitly. Irreducibility of the standard representation is R3.

P06 Schur's lemma (2): scalar commutants

Source. Schur 1.7(2), p. 7.

Target.

- A `FinIrreducibleRepres` with the explicit additional hypothesis that every `IntertwiningMap E E` is a scalar multiple of the identity.
- A proof that one-dimensional representations are split irreducible.
- A proof that the standard representation of S_3 is split irreducible over any commutative `LocalRing` in which 6 is invertible; L21 needs that inverse anyway. A 2×2 matrix commuting with $\rho(12)$

and $\rho(123)$ is scalar, by solving linear equations. Every division is meant to be by a divisor of 6; confirm this when the equations are written out. The representation has rank $2 > 0$, which L21 requires. Irreducibility then follows from L21. Instantiate at `RatField` (R7) and at `ComplexField` (R8a).

Relation: `restricted`. The book's eigenvalue argument needs $\mathbb{C}$ to be algebraically closed. Here the scalar commutant is proved per example; the general version is the M2 bridge.

P07 Uniqueness of multiplicities

Source. Prop. 1.8, p. 7.

Target. Take pairwise non-isomorphic split irreducibles V_i and two decompositions $V \cong \bigoplus V_i^{a_i} \cong \bigoplus V_i^{b_i}$, given as iterated `ProductLRepres`. Then $a = b$. Relation: `restricted`; it assumes characteristic 0 through injectivity of `natCoef` (L16).

Route (deviation).

- Characters are additive (P09) and the irreducible characters are orthonormal (P12). Together these give $\langle \chi_{V_i}, \chi_V \rangle = a_i = b_i$ in K; L16 then gives equality in $\mathbb{N}$.
- The book's route, through Hom-dimensions, needs additivity of $\dim \mathrm{Hom}_G$ over direct sums. The character route reuses work the pilot does anyway.

Dependencies: P09, P12, L16. Through P12, P07 depends on the Exercise 1.2 hole, so its status is `conditional-on-exercises` until that hole is closed; the characteristic-0 restriction appears in its signature.

P08 Basis-independent character; class function; $\chi(1) = \dim$

Source. §2.1, p. 13.

Target.

- `character (E : FinLinRepres R G) : G -> R`, defined through L05.
- `character-char`: it equals `Character E bv` for every basis.
- `character_conj`: `character E (h * g * inverse h) = character E g`.
- `character_ide`: the value at 1 is `natCoef (dimension E)`.

Relation: `same`.

Route.

- Class invariance: L02 followed by `Trace-prod`.
- The value at 1: `Character.of-1` (existing) and `dimension-unique`.

P09 Characters of sums, tensors and duals

Source. Prop. 2.1, p. 13.

Target.

- $\chi_{V \oplus W} = \chi_V + \chi_W$, using L06 and L07.
- $\chi_{V \otimes W} = \chi_V \chi_W$, using L12 and L14.
- $\chi_{V^*}(g) = \chi_V(g^{-1})$ and $\chi_{\mathrm{Hom}(V,W)}(g) = \chi_V(g^{-1})\chi_W(g)$, using L11, L13 and `Trace-transpose`.

The exterior-square conjunct is a registered pending item, not a hole. Relation: `reformulated` for the dual. The book writes $\overline{\chi_V}$; over $\mathbb{C}$ this agrees with $\chi_V(g^{-1})$ only after the roots-of-unity argument, which is deferred.

P10 Permutation characters count fixed points; Exercise 2.5 (hole)

Target.

```
-- | Fulton--Harris Exercise 2.5 (the original fixed-point formula).
\lemma permCharacter {R : CRing} {G : Group} (X : FinGSet G) (g : G)
  : Character (PermRepr X) (deltaBasis X) g
      = R.natCoef (FinSet.card (FixedPointsFin X g))
  => {?FH_L02_Ex2_5}
```

- The supporting definitions are `deltaBasis` (the indicator functions) and `FixedPointsFin` (the decidable subset).
- Both definitions are proved. Only the formula itself is the hole.

Relation: `same`. Consumers are R5 and the character of the regular representation (M2).

P11 Averaging onto invariants; $\dim V^G$ as a trace

Source. Prop. 2.8 and (2.9), pp. 15–16.

Target.

- `averageMap` $\varphi = |G|^{-1}\sum_g \rho(g)$, over `CRing`.
- φ is equivariant and idempotent, and its image is exactly the set of invariant vectors.
- Over `DiscreteField`: `natCoef (dimension (invariants E))` equals $|G|^{-1}\sum_g \chi_E(g)$.

Relation: `same` for Prop. 2.8. For (2.9) it is `reformulated`: the equation holds in K, and equality of natural numbers needs L16.

Route.

- Use the `FinSum` of `toLinearMap g` in `LinearMapAbGroup`, reindexed with `FinSum_Equiv` and `Group.translate-is-Equiv`.
- Trace commutes with the sum by L03 and the linearity of `Trace`.
- Dimension equals trace by L15.

P12 Orthogonality of split irreducible characters

Source. Thm. 2.12 and (2.10), p. 16.

Target. Two declarations, for split irreducibles V, W:

- `orthogonality-iso`: from `ModuleIso` intertwining $V \cong W$, conclude $\langle \chi_V, \chi_W \rangle = 1$.
- `orthogonality-noniso`: from `Not` (such an isomorphism), conclude $\langle \chi_V, \chi_W \rangle = 0$.

Relation: `restricted`. The input is split irreducibles, and the conclusion splits into two cases rather than a decided dichotomy.

Route.

- Apply P11 to `HomLRepres V W`. The invariants are the intertwiners by Exercise 1.2 (hole), so this item is `conditional-on-exercises`.

- In the non-isomorphic case, P03 forces every intertwiner to be zero, so the dimension is 0.
- In the isomorphic case, P06 gives the intertwiners a basis $\{\mathrm{id}\}$, transported along the isomorphism, so the dimension is 1.
- The character of `HomLRepres` comes from P09.

6 The exact S_3 regression suite

All objects are stated over an abstract commutative ring or `LocalRing` and instantiated at `RatField` and at `ComplexField`. The group is $G =$ `SymFinGroup 3` (L17). The generators are $\sigma = (1\,2)$ (`transposition`) and $\tau = (1\,2\,3)$ (`cycle`, `Symmetric.ard:139`). The sign is `signHom` (`Symmetric.ard:574`).

ID	Statement	Route
R1	The six permutations are explicitly enumerated and proved pairwise distinct and exhaustive. The implementation gives an equivalence $\mathrm{Fin}\,6 \simeq S_3$ and a formula for summing over that enumeration. Conjugacy-class sizes are not formalized separately.	L17; explicit finite cases
R2	Representations: trivial; sign ($g \cdot x = \mathrm{sgn}(g)x$, a new `CharacterLRepres` for a `MonoidHom` into the multiplicative monoid, using `signHom` and `sign_*`); `PermRepr` on `Fin 3`; standard ($e_1 - e_2$, $e_2 - e_3$, as a `BasedSplitSubLRepres` of `PermRepr`). The standard representation is given by an explicit inclusion $R^2 \to R^3$ and an explicit retraction, so it needs only the D3 definitions (wave 1), not the P01 bridge or the product basis L07.	D3, L18
R3	The standard representation extends to `FinIrreducibleRepres`, by L21 and the scalar commutant R7. Averaging any split invariant submodule gives an idempotent scalar c, and locality gives $c = 0$ or $c = 1$. This same argument applies over constructive $\mathbb{C}$.	R7, L21, P04a
R4	`PermRepr` is isomorphic to trivial $\oplus$ standard (P05).	P04b or explicit maps
R5	Character rows: trivial $(1, 1, 1)$, sign $(1, -1, 1)$, standard $(2, 0, -1)$ and permutation $(3, 1, 0)$ on identity, transposition and three-cycle representatives. The direct trace proofs do not use Exercise 2.5.	Explicit coordinate traces
R6	All nine pairings $\langle \chi_i, \chi_j \rangle$ are computed by summing over the six enumerated elements, with an explicit inverse of 6. They are 1 on the diagonal and 0 off it.	R1, L16; finite sums
R7	Every equivariant endomorphism of the standard representation is scalar, over any commutative ring. Comparing the coordinates for a transposition and a three-cycle proves this without division. Instantiate at `RatField` and `ComplexField`.	Explicit linear equations
R8a	**Constructive-$\mathbb{C}$ stage.** Instantiate the generic representation formulas, decomposition maps and scalar commutant at `ComplexField`. Supply the inverses of 3 and 6 through $\mathbb{Q} \to \mathbb{C}$. Prove complex irreducibility through L21; no general transfer of irreducibility from $\mathbb{Q}$ is assumed.	L21, L22, P04a, R4, R7
R8b	**Constructive-$\mathbb{C}$ characters.** Instantiate the generic character values and all nine finite-sum pairings at `ComplexField`, supplying the inverse of 6.	R5, R6, L22

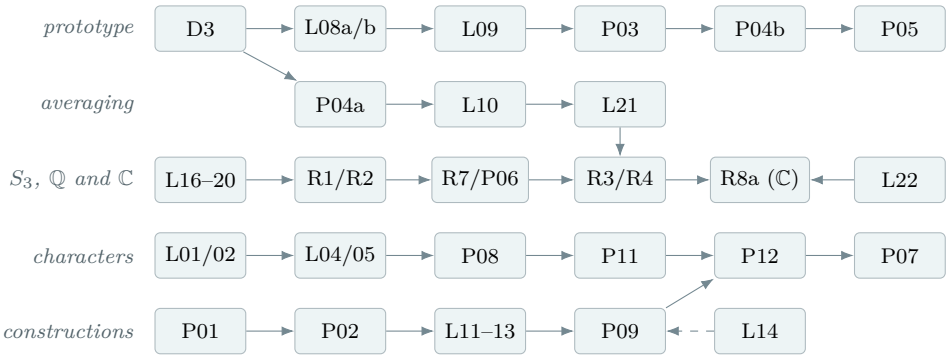
Arithmetic risk; abstract-then-instantiate is mandatory.

- `Rat` values are normalized by gcd, so closed `Rat` identities may not reduce to `idp`.
- Concrete `Complex` arithmetic unfolds into `re`/`im` components. `equation.cRing` then fails even on `x * y = y * x`, and `rewrite` cannot match under `*` (failure mode 16 of the `arend-prove` skill).
- Therefore R2–R8b are stated and proved over an abstract `{R : CRing}` or `LocalRing`, with explicit inverses of the needed integers as parameters. They are then instantiated twice, at `RatField` and at `ComplexField`.
- Concrete-field proofs are limited to supplying those inverses (L16, L22) and to the instantiation step.

7 Dependency order and schedule

The work runs in waves. Items inside a wave are *not* all independent: each wave lists its items as chains, written $a \to b$, and $a \mid b$ separates items that can run in parallel. An item may only start once everything before it in its chain, and every earlier wave it depends on, has checked.

The recommended start is a typechecked prototype of D3, L08 and Schur (wave 1). Wave 2 then tests the interfaces on the S_3 decomposition and scalar commutant, over both $\mathbb{Q}$ and constructive $\mathbb{C}$, before the general character theory is built.



The diagram shows the main chains; the item specifications list every dependency. P12 also depends on P03 and on R7/P06, and R8b (the $\mathbb{C}$ character stage) follows R5 and R6; these edges are not drawn. The dashed arrow marks L14, which feeds only the tensor conjunct of P09 and runs as a separate track.

Wave	Items	Gate to the next wave
0	Finish M0 (Section 1) | control tests | manifest skeleton for all P/L/R items | statement review of D3, L08a, L08b, P03	D3 statement review signed off
1	**Prototype:** D3 definitions (`SplitSubLRepres`, `FinIrreducibleRepres` over `NonZeroRing`, old notion renamed) → (L08a | L08b | L09) → P03; everything typechecked with no goals	Schur typechecks; D3 stands or is revised
2	**Interface test on S_3 over $\mathbb{Q}$ and $\mathbb{C}$.** Chain A: (L16 | L17 | L18 | L19 | L20) → R1 → R2 → P06/R7. Chain B, in parallel with A: P04a → L10 → L21 (generic; no S_3 input). Chain C, after A and B: R3 → R4 (with explicit decomposition maps, since P04b comes in wave 3). Chain D, after C: L22 → R8a	Decomposition, commutant and irreducibility check over both fields, with no zero test on $\mathbb{C}$
3	L01 → ((L02 → L04 → L05 → P08) | (P01 bridge with $\rho(g^{-1})$, migration → P02 with hole)); in parallel: P01b | (P04b → P05) | L03	Characters and Maschke complements check
4	(L06 | L07 | P10 with hole) → R5 → R8b-values; (L11–L13 | L15) → P11	Invariant-dimension formula and $\mathbb{C}$ character values check
5	P09 (sum, dual, Hom conjuncts) → P12 → (P07 | R6 → R8b-pairings)	**Core pilot complete** (Section 2)
Track T	L14 → tensor conjunct of P09, from wave 1 onwards	**All selected items complete**

8 Verification protocol

Edit loop. Fast checks

Run checks from `arend-lib/`, through the daemon: `arend -serialize <Module>[:<def>]`. Look up names with `-ss`, `-ps`, `-fu` and `-ch` before writing any helper. After a class-level change, the daemon has to be refreshed with `arend -daemon-refresh`. The `arend-formalize` and `arend-prove` skills describe this loop.

Acceptance. Fresh checks

```
cd arend-lib
arend --no-daemon -r -c Algebra.Group.Representation.Irreducible \
  Algebra.Group.Representation.SplitSubLRepres Algebra.Group.Representation.Hom \
  Algebra.Group.Representation.Characters Algebra.Group.Representation.MaschkeLemma \
  Algebra.Group.Representation.Invariants ...           # every touched module
arend --no-daemon -t -r -c                             # library tests, incl. S3Test
```

The acceptance script parses the `[GOAL]` and `[ERROR]` lines.

- The run passes only if the goal set is exactly {`FH_L01_Ex1_2`, `FH_L02_Ex2_5`} at the manifest's declarations, and there are no errors.
- Goals alone do not fail the exit code, so the script has to check the goal set itself.
- Under `-c`, the double checker reports goals like errors. Those lines are matched against the manifest too.

Controls. Prove that the harness can fail

Two tests go under `test/Algebra/Representation/`.

1. A deliberately wrong proof of `0 = {Nat} 1`, wrapped in the `fails` meta. The control is that removing `fails` must turn the run red.
2. A small module that imports `Logic.Classical` and uses `alem`. The axiom audit must flag it.

Axiom audit. Two levels

- **Level 1, the gate.** The import-closure script reports any path to `Logic.Classical` or `Logic.Choice`. It over-approximates real dependency, so a "no" is conclusive.
- **Level 2, investigation.** If Level 1 ever says "yes", `-fu alem`, `-fu achoice`, `-fu acchoice` and `-fu adchoice` locate the actual uses.
- **Hypothesis review.** Reviewers also check signatures for classical strength: any hypothesis of the shape `\Pi (P : \Prop ) -> ...` or a decision procedure for an undecidable property.

Semantic review. Per item

- Check that the formal statement matches the source in quantifiers, scalar class, finiteness, and isomorphism versus equality.
- Record the relation and the data that replaces each classical step.
- Run the Phase-4 finish checklist of the `arend-formalize` skill: goals, names, comments, placement, hypothesis weakening, redundancy and layout.
- Try weakening each hypothesis, in particular `DiscreteField` to `Field` or `CRing`, and record which ones the checker accepts. Each successful weakening is a better statement.

9 Manifest records

Location. One JSON Lines file, `arend-lib/docs/fulton-harris/manifest.jsonl`, with one record per P-, L- and R-item. It uses the whole-book plan's schema and states. Two representative records, shown in YAML for readability:

```
id: FH.L01.Schur1.7.1            # P03
kind: theorem
source: {printed_page: 7, pdf_page: 22, label: "Schur's Lemma 1.7 (1)"}
original: "A G-map between irreducibles is zero or an isomorphism."
constructive: "For split-irreducible based representations over a discrete field,
               an intertwiner is zero or a module isomorphism."
source_relation: reformulated
formal_target: Algebra.Group.Representation.Irreducible:SchurLemma
dependencies: [GAP.L08.smithSplit, GAP.L09.splitSurj, KernelLRepres, ImageLRepres]
status: planned
statement_review: pending
axiom_policy: no_excluded_middle_or_choice

id: GAP.L15.idempotentTrace
kind: infrastructure
statement: "endoTrace p = natCoef (dimension (image-fin p)) for p idempotent"
formal_target: Algebra.Linear.VectorSpace:idempotent-trace   # placement to review
consumers: [FH.L02.Prop2.8, FH.L02.Thm2.12]
acceptance: "checked with no goals; no new hypotheses beyond DiscreteField"
status: planned
```

Status after a check. Once an item is checked, its record gains the jar hash, the source revision, the command, the exit status, the goal lines and the reviewer outcome.

10 Measurement, risks and decisions needed

Measurement. What to record per item

- Task class: wrapper, routine algebra, bridge, or major theorem.
- Repair cycles used, out of a budget of five.
- Wall-clock checking time.
- Human review minutes.
- Lines added and lines of reused library code.
- The outcome classification: success, source ambiguity, missing lemma, elaboration failure, or search failure.

After wave 2, re-estimate the remaining waves from the observed medians. The pilot's purpose is calibration, so these records are part of the deliverable.

Risk	Early signal	Response
D3 too weak for Schur	The wave-1 prototype cannot state L08a/L08b cleanly, or P03 needs extra hypotheses	Revise D3 before wave 2; candidates: projective (finitely generated, split) instead of free submodules
Complex bridge stalls	R8a or R8b needs a zero test, or concrete `Complex` terms block unification	Keep every S_3 statement abstract (Section 6); supply only inverses of $2, 3, 6$ through L22; record any remaining step as a named gap, never as a hidden decision procedure
Truncation friction	`character` or `endoTrace` cannot eliminate `isFinModule`	Follow `dimension-pair` exactly; fall back to `BasedLinRepres`-only statements for the pilot
Consumers of holes do not elaborate	Errors at use sites of `invariantHom`	Use the `-cond` hypothesis form (D6)
Rational arithmetic is slow or stuck	R5/R6 time out or `equation` fails on literals	Abstract-field statements, instantiate at the end (Section 6)
L14 (tensor basis) overruns	No progress after one wave	Keep it as `pending`: the pilot can reach "core complete" but not "all selected items complete" (Section 2); P12 is unaffected
Order conventions	Three library conventions compose diagrammatically: `toMatrix_*` (row convention), `LinearMapRing` (`f * g = g ∘ f`) and `SymmetricGroup` (`(e * e') j = e' (e j)`). As a consequence, $g \mapsto \rho(g)$ is an antihomomorphism into `LinearMapRing`	Name lemmas by mathematical order; add a regression test that fixes each convention
Daemon flag locking	A "fresh" run silently reuses caches	Acceptance uses `-no-daemon` only

Decisions. Needed from the project owner before wave 1

1. Approve D3 as the *pilot* notion `FinIrreducibleRepres`, over any `NonZeroRing`, together with its recorded restrictions and the three open questions listed under it.
2. Confirm that the name `IsIrreducible` stays reserved for the general-module design D3b, and approve renaming the old notion to `IsStronglyIrreducible` instead of deleting it.
3. Choose the bridge direction for P01: $g \mapsto \rho(g^{-1})$, or a homomorphism from the opposite group.
4. Approve the character route for P07, with its characteristic-0 restriction and its status `conditional-on-exercises`.
5. Manifest location resolved: `arend-lib/docs/fulton-harris/`.

Revision note. Corrections in this version

This version incorporates reviews of the first draft and the hierarchy refactor of 20 October 2026. The current API uses a descendant class; the earlier standalone `IsFinIrreducible` predicate has been removed.

- L08 is split into L08a and L08b. Image splitting is false over a general `SmithDomain`.
- D3 is now a pilot-only notion with no decidable equality; the general design is D3b.
- A constructive-$\mathbb{C}$ bridge for S_3 (R8a, R8b, L22) is added.
- The direction of the P01 homomorphism is corrected, and L02 is no longer called a monoid map.
- R3 now goes through the scalar commutant and Maschke averaging (L21).
- The exit criteria now have two tiers.
- Second review: L21 requires positive rank. Its proof uses `LocalRing.connected`. The scalar ring must be a `NonZeroRing`, which excludes the zero ring; waves list their internal dependency chains; L19 moves to wave 2; R8 is split into a structural stage (R8a) and a character stage (R8b); and the claim about complex kernels and images is softened.
- Third review: $0 \neq 1$ is described as a nontriviality assumption; stale "any `Ring`" wording is corrected; R2 depends only on D3 and L18.
- Wave 1: the old notion is named `IsStronglyIrreducible`. Its lemma is `SchurLemma-strong`. A nested name `IsIrreducible.Strong` would need a definition called `IsIrreducible`, and that name is reserved for D3b. The new Schur lemma concludes `IsEquiv f`, as the old one did.
- **Still deferred:** D3b, the general-module replacement for irreducibility. Completing this pilot validates finite-dimensional irreducibility only.

11 Implementation status (updated 20 October 2026)

Class refactor (20 October). The finite representation class now lives in `Group/Representation/Finite.ard` and inherits the existing `FinModule` structure. The irreducible descendant in `Group/Representation/Irreducible.ard` has named `nontrivial` and `irreducible` fields. The lemma `basis-nonempty` derives nonemptiness of the basis. Constructors return class extensions of the specified action; Schur, orthogonality and multiplicity theorems consume those classes directly. The former `isFinRepres` field and `toFinModule` adapter are removed: a finite representation is already a finite module by inheritance.

The subobject interfaces now form one hierarchy. `SplitSubLModule` in `Module/SubLModule/Split.ard` extends the existing `SubLModule`; `SplitSubLRepres` extends it and the existing `SubLRepres`. `BasedSplitSubLModule` in `Linear/BasedSplitSubLModule.ard` and `BasedSplitSubLRepres` add the coordinate data used by kernel/image algorithms and the explicit S_3 plane. The old pilot module `Linear/SplitSubLModule.ard` is replaced, and all imports are migrated. Both split classes were introduced by the pilot; they are not pre-pilot library interfaces. Only the membership parents were already present. Averaging, complements, split characters and irreducibility consume the basis-free descendant. Dimension formulas explicitly consume the based one. The proofs `imageEquiv`, `splitMono-dichotomy` and `splitMono-irreducible` establish compatibility with arbitrary split intertwiner presentations; `coordinateIso`, `arrayToSubEquiv` and `splitCharacter-coordinates` establish compatibility with coordinates. The irreducibility premise is unchanged by this hierarchy refactor.

All 466 library modules pass a fresh normal typecheck. The pilot's 26 manifest modules pass fresh double-checking, and 136 formal targets are verified. All 704 tests pass. Exactly the two registered exercise holes remain. The F2 companion also passes a fresh double-check. A separate whole-library double-check encounters the previously recorded error at `Category/Limit.ard:1034`; it is not a pilot-module failure.

All P-, L- and R-items were implemented in `arend-lib`. The only open goals in the library outside block comments are the two registered exercise holes. Paths are relative to `arend-lib/src/Algebra/`.

Item	Declarations (module)	Notes
D3	`SplitSubLRepres`, `SplitSubLRepres.fromInvariant`, `FinIrreducibleRepres` (`Group/Representation/SplitSubLRepres`, `Irreducible`)	Old notion renamed `IsStronglyIrreducible`, Schur as `SchurLemma-strong`. `FinIrreducibleRepres` now extends `FinLinRepres`; the latter inherits `FinModule`. The named fields are `nontrivial` and `irreducible`; nonemptiness of the basis is derived by `basis-nonempty`. `SplitSubLRepres` descends from the existing `SubLRepres` and the new `SplitSubLModule`; the based descendants add coordinates. The split-monomorphism bridges preserve the quantification domain.
P01	`toUnitsHom`, `fromUnitsHom` and both round-trip laws (`Hom`); `PermRepr-not-FinIrreducible` (`PermutationRepresentation`); `rank-one-idempotent`, `rank-one-irreducible` (`Irreducible`)	Uses `InvGroup` from `Arith/Nat/EulerTotient`. Rank-one irreducibility holds over every `NonZeroCRing` without nontrivial idempotents (`IsConnected`), without averaging.
P01b	`IsStronglyIrreducible.excludedMiddle`	Proved; no axiom.
P02	`HomLRepres`, `DualLRepres`, `invariantHom`	`invariantHom` is the hole `FH_L01_Ex1_2`.
P03	`SchurLemma`	Over `DiscreteField`; concludes `IsEquiv f`.

Item	Declarations (module)	Notes
P04	`splitProjector`, `maschke-split` (P04a, `Averaging`); `maschke-complement` (P04b, `Complement`)	P04a over `CRing`; `maschke-split` makes `incl` a `SplitMono` in `RepresentationCat`; P04b weakened during cleanup from `DiscreteField` to `SmithDomain`.
P05	`perm3-complete-reducibility` (`Symmetric3`)	Restricted to the presented S_3 input.
P06	`std-commutant`	Holds over every `CRing`: the equations need no division (the plan expected divisors of 6).
P07	`multiplicities` (`Multiplicities`)	Character route; characteristic 0 as a hypothesis; conditional on Ex. 1.2.
P08	`character`, `character-char`, `character_conj`, `character_ide` (`Characters`)	Basis independence via L04.
P09	`character_sum`, `character_hom`, `character_dual` (`CharacterFormulas`), `character_tensor` (`Tensor`)	Exterior square still deferred.
P10	`permCharacter`	Hole `FH_L02_Ex2_5`; no consumers.
P11	`invProj`, `invProj.trace`, `dim-invariants`, `invariants-split` (`Invariants`)	Dimension formula over `CRing`; existence over `DiscreteField`.
P12	`orthogonality-noniso`, `orthogonality-self`, `orthogonality-iso` (`Orthogonality`)	Conditional on Ex. 1.2.
L01–L05	`toLinearMap_*`, `toMatrix-action_*`, `Trace-coords`, `Trace-basis-indep`, `endoTrace` (`Module/TraceCoords`)	L04 holds over any `CRing`, for bases of possibly different lengths, with no inverses. L03 is subsumed by `Trace-coords`.
L06–L07, L11–L14	`CoordSystem`, `productCS`, `homCS`, `tensorCS`, `deltaCS`, `arrayCS` (`Module/CoordSystem`, `Tensor`)	One mechanism (finite set-indexed coordinates and `Trace-coords`) replaces block and Kronecker matrices.
L08–L09	`kernel-split`, `image-split`, `split-full-surj` (`Linear/BasedSplitSubLModule`)	As corrected: kernel over `SmithDomain`, image over `DiscreteField`.
L10	`split-dim-<=` (`Linear/BasedSplitSubLModule`)	No longer used downstream: rank-one irreducibility goes through `rank-one-idempotent`.
L15	`trace-split`	Cyclicity of the trace; no kernel basis.
L16, L22	`natCoef-Inv-Rat`, `ratToComplex`, `natCoef-Inv-Complex` (`../Arith/Complex/RatHom`)	
L17–L20	`SymFinGroup`, `SymAction` (`Group/Symmetric/Action`), `FixedPointsFin`, `right-translate`	L20's inversion reindexing was not needed and was removed.
L21	`irreducible-from-commutant` (`Averaging`)	Uses `LocalRing.connected`.
R1–R7	`Symmetric3`, `Symmetric3Characters`	R1 is an explicit enumeration $\mathrm{Fin}\,6 \simeq S_3$; class sizes are not stated separately, since the sums run over all six elements. R6 covers all nine pairings, the reversed ones via the general lemma `pairing-symm`. Trivial and sign are irreducible over any `NonZeroCRing` without nontrivial idempotents (tested over $\mathbb{Z}$, also for $C_2 = S_2$).
R8a/b	`test/Algebra/Representation/S3Test.ard`	Irreducibility, decomposition, character values and pairings over `ComplexField`, with the only inverse being the image of 1/6.

Conditional declarations. Through `invariant-intertwining`, these depend on the hole `FH_L01_Ex1_2`:

- `orthogonality-noniso`, `orthogonality-self` and `orthogonality-iso`;
- `delta-case`, `count-pairing` and `multiplicities`.

Nothing consumes `FH_L02_Ex2_5`. The S_3 results R1–R8 do not depend on either hole.

Cleanup pass.

- 87 unused imports were removed. Each removal was kept only if the module still typechecked with the same goals.
- Dead helpers were deleted.
- Plan IDs in doc comments were replaced by book references.
- Hypothesis weakening was tried; P04b now holds over any `SmithDomain`.
- Not attempted: removing redundant explicit implicit arguments; weakening `invariants-split` using a kernel splitting.

Completion audit.

- The manifest is `arend-lib/docs/fulton-harris/manifest.jsonl`: 48 records with formal targets, source relation and status. This settles decision 5.
- `arend-lib/tools/autoformalization/pilot_acceptance.py` checks in turn:
  1. every target exists;
  2. a fresh `-no-daemon -r -c` run, with targets in dependency order, reports no errors;
  3. the goals reported are exactly the registered holes;
  4. the test suite passes;
  5. the import-closure axiom audit is clean;
  6. the declaration-level consumer closure of the holes, computed with `-fu -json`, matches the manifest statuses;
  7. a negative control: the classical fixture is flagged;
  8. a negative control: a temporary module with an invalid proof is rejected;
  9. doc comments are well formed: multi-line docs use `{- | ... -}`, not a `-- |` line followed by `--` continuation lines, which would fall outside the documentation. Block docs must not contain `{-` or `-}`.

  Check 1 queries the symbol index rather than the file text, so a definition swallowed by a stray comment delimiter is caught.
- A third control, inside the test suite, uses `fails`.
- Run on 12 October: `ACCEPTED`. A mutation of the manifest (marking P12 as `checked`) is rejected.
- Targets are checked in dependency order because, in a multi-module run, the CLI does not print the goals of a module first typechecked as a dependency.

Not yet done.

- The general-module design D3b.
- The general complex irreducibility-to-commutant bridge in F2 below. The F2 design decision and standalone averaging prototypes are recorded in the 19 October update; they are not integrated into arend-lib. F1 is implemented.

Follow-up items for Lectures 3 onward

Relation of the two irreducibility notions. Before the basis-free redesign (which predates the current hierarchy refactor), `SplitSubLRepres` tested only free summands R^k with a linear retraction. The current notion tests every split subrepresentation (`SplitMono` in `LModuleCat`), and it is strictly stronger:

- **New implies old** over every `NonZeroRing`, via `fromInvariant`: a zero subrepresentation R^k has $k = 0$.
- **Old does not imply new** in general. Over $R = \mathbb{Z} \times \mathbb{Z}$, the trivial representation on R is irreducible in the old sense, but $R \times 0$ is a nonzero, non-surjective split subrepresentation. Hence `rank-one-irreducible` needs `IsConnected`.
- **They agree** over a `DiscreteField` (split the image of the idempotent $incl \circ retr$ by `image-split`). They also agree over a `LocalCRing` if finitely generated projective modules over local rings are free. That theorem is constructive in the literature (Lombardi–Quitté; exact statement to be checked), but it is not in arend-lib.
- Neither equivalence is formalized.

Over $\mathbb{C}$ the new notion is the more usable one. `ComplexField` is not a `DiscreteField`, so a basis of the image of an idempotent is not available. Isotypic projections, the projection formula (§2.4) and the decomposition of the regular representation all produce such images, and they are `SplitSubLRepres` values directly, with π as retraction.

F1 Basis-free character of a split subrepresentation

- **Problem.** `character` is defined for `FinLinRepres`, which needs a truncated basis. The summand `S.sub` of a `SplitSubLRepres` has none. Cor. 2.15 ($\langle \chi, \chi \rangle = 1$ implies irreducible), multiplicity counts and dimension counts need the characters of summands.
- **Proposal.** Define $\chi_S(g) = \mathrm{tr}_E(incl \circ g \circ retr)$ using `endoTrace` on E. Prove that it agrees with `character` when `S.sub` has a basis (trace cyclicity, as in L15). Prove additivity $\chi_E = \chi_S + \chi_C$ for the decomposition of `maschke-complement`.
- **Alternative.** Formalize "finitely generated projective modules over a `LocalCRing` are free". This would also close the gap between the two irreducibility notions, but it costs more.
- **Base.** Any `CRing`; no division needed.
- **Implemented** (`Group/Representation/SplitCharacter`, `Module/TraceCoords`):
  - `endoTrace-comm` and `endoTrace-sum`: cyclicity and additivity of the basis-free trace;
  - `splitCharacter`, defined with the inherited retraction, and `splitCharacter-retr`: any other retraction gives the same value;
  - `splitCharacter-basis`: if `S.sub` has a basis, the value is its ordinary `Character`. The proof expands `S.sub` along *retr* of a basis of E and applies `Trace-coords`, so no cyclicity across different modules is needed;
  - `splitCharacter-sum`: $\chi_E = \chi_S + \chi_C$ for any splitting $e = incl_S(q\, e) + incl_C(q'\, e)$ with retractions q, q'; and `maschke-character`: over a `SmithDomain` with $|G|$ invertible, every split S has a complement C with $\chi_E = \chi_S + \chi_C$;
  - test `std-splitCharacter-Q`: for the standard summand of the S_3 permutation representation over $\mathbb{Q}$, the split character equals `character (StdFin RatField)`.

  Manifest record `GAP.F1`; acceptance run: `ACCEPTED`, 684 tests.

F2 Schur's lemma over $\mathbb{C}$

- **Problem.** `SchurLemma` is proved over `DiscreteField`, using `kernel-split` and `image-split`. Over `ComplexField`, the kernel and image of an intertwiner are not known to be split, so the lemma does not apply. The pilot avoided this over $\mathbb{C}$: it used the commutant route (L21) and explicit S_3 computations. From Lecture 3 on, Schur over $\mathbb{C}$ is used throughout.
- **This holds for both irreducibility notions.** It is not caused by the redesign.
- **Decision (20 October 2026).** Keep the existing `DiscreteField` dichotomy. For constructive complex character theory, use the trace of averaging with explicit scalar-commutant and zero-Hom hypotheses. Deriving those hypotheses from `FinIrreducibleRepres` is a separate, still open theorem. The prototypes below verify this separation.

F2 update: the scalar obstruction

The obstruction is already visible in dimension one, independently of any kernel algorithm. Let R be a commutative local ring and let $E = R$ carry the trivial action of any group. The existing `rank-one-irreducible` proves an extension of E to `FinIrreducibleRepres`, using `LocalRing.connected`. For any $c \in R$, the map $m_c(x) = cx$ is an intertwiner. A proposed general Schur dichotomy gives

$$m_c = 0 \ \vee \ m_c \text{ is an isomorphism} \quad \Longrightarrow \quad c = 0 \ \vee \ c \text{ is a unit.}$$

In the first case evaluate at 1. In the second case take the inverse image of 1; if it is b, then $cb = 1$, so c is invertible by commutativity. For a `Field`, this constructs a `DiscreteField` on the same carrier and operations. This is an implication from the proposed dichotomy, not a proof of its negation inside Arend. It identifies the additional assumption that a purported constructive proof would supply.

The distinction between a field where noninvertibility implies zero and one where every element is zero or invertible is also discussed by Blechschmidt–Schuster, *Maximal ideals in countable rings, constructively*. The reduction above is checked against this repository's definitions, without using a classical axiom or any exercise hole.

Consequently, apartness-based elimination can be useful with explicit rank or splitting evidence, but it cannot supply the unrestricted dichotomy over `ComplexField` without also supplying scalar discreteness. Do not replace positive apartness by logical negation of equality in such hypotheses.

F2 update: averaging without a basis of the invariants

For finite free V, W over a commutative ring with $|G|$ invertible, put $H = \mathrm{Hom}_R(V, W)$ and let P be the existing averaging projection on H. The existing `invProj.trace` and `character_hom` give

$$\langle \chi_V, \chi_W \rangle = \mathrm{tr}_H(P).$$

This identity does not require `invariants-split`. It supports two constructive consequences:

- If every intertwiner $V \to W$ is zero, every $P(f)$ is zero, hence $\mathrm{tr}_H(P) = 0$. This assumes zero Hom explicitly; it does not deduce it from non-isomorphism of irreducibles over $\mathbb{C}$.
- Suppose V has a basis with a specified index j, and every intertwiner $V \to V$ is scalar. Define $\lambda(f)$ to be the j-th coordinate of $f(v_j)$. Then $\lambda(I) = 1$ and $P(f) = \lambda(P(f))I$. The rank-one trace identity $\mathrm{tr}(u \mapsto d(u)v) = d(v)$ gives $\mathrm{tr}_H(P) = \lambda(P(I)) = \lambda(I) = 1$.

Both consequences hold over `CRing`; no zero test, invariant-submodule basis, or inverse of $\dim V$ is used. A selected basis index witnesses positive rank for the second statement. The scalar-commutant

assumption is retained explicitly even though it also implies irreducibility over a local ring by the existing `irreducible-from-commutant`.

Standalone formal checks. The companion library is `docs/fulton-harris-f2`, with dependency `arend-lib`:

- `src/SchurObstruction.ard`: `SchurDichotomy`, `zero-or-invertible-of-schur`, and `discrete-from-schur`.
- `src/AveragingRoute.ard`: `pairing-average`, `trace-rank-one`, `orthogonality-zero-hom`, and `orthogonality-self-commutant`.
- `src/ComplexCheck.ard`: `std-self-pairing-C` instantiates the latter theorem with `std-commutant` and `ComplexField`, using only the inverse of 6 for averaging.

The two new orthogonality results and their complex specialization are *conditional on Exercise 1.2*, through the existing `invariant-intertwining`. The scalar obstruction, rank-one trace lemma, and averaging trace identity do not consume that hole. No new exercise holes are introduced. These are standalone checks, not additions to the production manifest or a replacement for the existing hole-independent S_3 computations.

From `docs/fulton-harris-f2`, run:

```
arend --no-daemon -r -c -L ../.. \
  Algebra.Group.Representation.Hom \
  Algebra.Group.Representation.CharacterFormulas \
  SchurObstruction AveragingRoute ComplexCheck
```

List the two exercise-bearing dependency modules first so their registered goals remain visible. A successful check must have no errors and exactly the existing goals `FH_L01_Ex1_2` and `FH_L02_Ex2_5`. On 20 October, after the hierarchy migration, this command exited 0 with exactly those two goals; the three prototype modules passed the double-check. Their import closures contain neither `Logic.Classical` nor `Logic.Choice`. The scoped result is recorded in `docs/fulton-harris-f2/validation.json`.

Remaining before general Lecture 3 claims. Integrate the reusable trace and averaging lemmas with the production manifest and its dependency audit. Then establish which additional hypotheses suffice to derive scalar commutants and zero Hom from the chosen complex irreducibility notion. An invariant positive-definite Hermitian form and spectral projections are a possible route to investigate, not an implemented theorem. Neither the prototype nor F1 establishes that bridge, complete reducibility over $\mathbb{C}$, or a general complex Schur lemma.